

Day 02 – Linux Commands Practice

Task

Today's goal is to **build your Linux command confidence**.

You will create a cheat sheet of commands focused on:

- Process management
- File system
- Networking troubleshooting

This is the command toolkit you will reuse for years.

File & Directory Navigation

`pwd` – Print the current working directory.

`ls` – List files and directories (`ls -la` shows hidden and detailed info).

`cd` – Change directory.

`mkdir` – Create a new directory.

`touch` – Create an empty file or update timestamps.

`cp` – Copy files or directories.

`mv` – Move or rename files.

`rm` – Remove files or directories (use with caution).

Viewing & Editing Files

`cat` – Display file contents.

`nano` / `vim` – Edit files directly in the terminal.

`head` / `tail` – Show the first or last lines of a file.

`grep` – Search for text patterns inside files.

System Monitoring & Management

`top` – Real-time CPU/memory usage.

`ps` – List running processes.

`kill` – Terminate a process by PID.

`df -h` – Show disk space usage.

`free -h` – Display memory usage.

`uptime` – Show system running time and load averages.

`du -sh <dir>` – Show size of a directory.

`mount` / `umount` – Attach or detach file systems.

`lsblk` – List block devices (disks, partitions).

`file <filename>` – Identify file type.

Networking Commands

ping – Test connectivity to a host.
ip a – Show network interfaces and IP addresses.
wget – Download files from the internet.
curl – Transfer data from/to URLs.
ssh – Securely connect to remote servers.

Permissions & Ownership

chmod – Change file permissions.
chown – Change file ownership.
whoami – Show current logged-in user.

Package Management (Ubuntu)

apt-get update – Refresh package lists.
apt-get install <package> – Install software.

File & Directory Management

mkdir <dir> – Create a new directory.
rmdir <dir> – Remove an empty directory.
touch <file> – Create an empty file or update timestamp.
cp <src> <dest> – Copy files or directories (-r for recursive).
mv <src> <dest> – Move or rename files.
rm <file> – Delete files (rm -r <dir> for directories)

Viewing & Searching

cat <file> – Display file contents.
less <file> – Scroll through file contents.
head <file> – Show first 10 lines.
tail <file> – Show last 10 lines (tail -f for live logs).
find <path> -name <file> – Search for files.
grep "pattern" <file> – Search text inside files.

Category	Common Commands	Purpose
Navigation	pwd, ls, cd	Move around filesystem

Category	Common Commands	Purpose
File Management	mkdir, touch, cp, mv, rm	Create, copy, move, delete
Viewing/Editing	cat, nano, vim, grep	Read & edit files
System Monitoring	top, ps, df, free	Check performance & resources
Networking	ping, ip, wget, curl, ssh	Connectivity & downloads
Permissions	chmod, chown, whoami	Manage access rights
Package Management	apt-get commands	Install/remove software

Snaps (few examples)

```
dash@ip-172-31-36-189:~$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                7:0      0  28.2M 1 loop /snap/amazon-ssm-agent/13009
loop1                7:1      0   74M 1 loop /snap/core22/2411
loop2                7:2      0  48.4M 1 loop /snap/snapd/26382
loop3                7:3      0  49.3M 1 loop /snap/snapd/26865
nvme0n1              259:0     0   10G 0 disk
├─nvme0n1p1          259:1     0   8.9G 0 part /
├─nvme0n1p13         259:2     0 1023M 0 part /boot
├─nvme0n1p14         259:3     0    4M 0 part
└─nvme0n1p15         259:4     0  106M 0 part /boot/efi
dash@ip-172-31-36-189:~$
```

```
dash@ip-172-31-36-189:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        8.6G  2.7G  5.9G  32% /
tmpfs            953M     0  953M   0% /dev/shm
tmpfs            382M  924K  381M   1% /run
efivarfs         128K   3.3K  120K   3% /sys/firmware/efi/efivars
tmpfs            953M     0  953M   0% /tmp
none             1.0M     0   1.0M   0% /run/credentials/systemd-journald.service
none             1.0M     0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M  163M  760M  18% /boot
/dev/nvme0n1p15 105M   6.3M   99M   7% /boot/efi
none             1.0M     0   1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M     0   1.0M   0% /run/credentials/getty@tty1.service
none             1.0M     0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            191M   8.0K  191M   1% /run/user/1000
dash@ip-172-31-36-189:~$
```

```
dash@ip-172-31-36-189:/scripts$ mkdir test
dash@ip-172-31-36-189:/scripts$ ll
total 40
drwxrwxrwx  4 dash dash 4096 Jun  6 11:34 ./
drwxr-xr-x 21 root root 4096 Jun  6 02:39 ../
drwxrwxr-x  7 dash dash 4096 Jun  6 09:34 Devops/
-rwxrwxrwx  1 dash dash  246 May 30 14:41 argument.sh*
-rwxrwxrwx  1 dash dash  161 Jun  1 03:46 createfile.sh*
-rwxrwxrwx  1 dash dash    0 May 30 16:21 dash.txt*
-rwxrwxrwx  1 dash dash  128 May 30 03:27 hello.sh*
-rwxrwxrwx  1 dash dash  195 Jun  1 05:24 install_package.sh*
drwxrwxr-x  2 dash dash 4096 Jun  6 11:34 test/
-rwxrwxrwx  1 dash dash  260 May 30 15:27 user_input.sh*
-rwxrwxrwx  1 dash dash  282 May 30 14:20 variable.sh*
dash@ip-172-31-36-189:/scripts$
```

```
dash@ip-172-31-36-189:/scripts$ rmdir test
dash@ip-172-31-36-189:/scripts$ ll
total 36
drwxrwxrwx  3 dash dash 4096 Jun  6 11:35 ./
drwxr-xr-x 21 root root 4096 Jun  6 02:39 ../
drwxrwxr-x  7 dash dash 4096 Jun  6 09:34 Devops/
-rwxrwxrwx  1 dash dash  246 May 30 14:41 argument.sh*
-rwxrwxrwx  1 dash dash  161 Jun  1 03:46 createfile.sh*
-rwxrwxrwx  1 dash dash    0 May 30 16:21 dash.txt*
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-rwxrwxrwx  1 dash dash  282 May 30 14:20 variable.sh*
dash@ip-172-31-36-189:/scripts$
```

```
ubuntu@ip-172-31-36-189:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
ubuntu@ip-172-31-36-189:~$
```

```
ubuntu@ip-172-31-36-189:~$
ubuntu@ip-172-31-36-189:~$ echo "I will become a SRE"
I will become a SRE
ubuntu@ip-172-31-36-189:~$
```

```
ubuntu@ip-172-31-36-189:~$
ubuntu@ip-172-31-36-189:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-36-189:~$
ubuntu@ip-172-31-36-189:~$
```

```
ubuntu@ip-172-31-36-189:~$ cd /
ubuntu@ip-172-31-36-189:/$
```

```
ubuntu@ip-172-31-36-189:/$ ls
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin snap srv sys tmp usr var
ubuntu@ip-172-31-36-189:/$
```

```
ubuntu@ip-172-31-36-189:/$
ubuntu@ip-172-31-36-189:/$ uptime
12:41:42 up 22:14,  3 users,  load average: 0.00, 0.00, 0.00
ubuntu@ip-172-31-36-189:/$
ubuntu@ip-172-31-36-189:/$
```